Limits & Derivatives

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: One-sided limits**

Using the same piecewise function, does $\lim_{x \to 2} f(x)$ exist? State its value.

Yes — both one-sided limits equal 3, so $\lim_{x \to 2} f(x) = 3$.

2 **[Foundational]** • **Target: The power rule**

Differentiate $f(x) = 3x^4 - 2x^2 + 7$.

$f'(x) = 12x^3 - 4x$.

3 **[Foundational]** • **Target: The product rule**

Differentiate $f(x) = x^2(3x + 1)$ using the product rule.

$f'(x) = (2x)(3x + 1) + x^2(3) = 6x^2 + 2x + 3x^2 = 9x^2 + 2x$.

4 **[Foundational]** • **Target: Tangent lines**

Find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.

$f'(x) = 2x$, so $f'(3) = 6$.

5 **[Foundational]** • **Target: Tangent lines**

Find the equation of the tangent line to $f(x) = x^2$ at $x = 2$.

$f(2) = 4$, $f'(2) = 4$. $y - 4 = 4(x - 2) \Rightarrow y = 4x - 4$.

6 **[Foundational]** • **Target: Tangent lines**

The graph of $f(x) = x^2 - 2x$ is shown. Find the equation of the tangent line at $x = 3$ (the marked point).

$f(3) = 9 - 6 = 3$. $f'(x) = 2x - 2$, so $f'(3) = 4$. $y - 3 = 4(x - 3) \Rightarrow y = 4x - 9$.

7 **[Foundational]** • **Target: Tangent lines**

Find the x-coordinate(s) where the tangent to $f(x) = x^3 - 3x$ is horizontal.

$f'(x) = 3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Increasing, decreasing, and the second derivative

For $f(x) = x^2 - 4x + 1$, find $f'(x)$ and determine the interval where f is increasing.

$f'(x) = 2x - 4$. Increasing when $f'(x) > 0$, i.e. $x > 2$.

9 [Intermediate] • Target: Increasing, decreasing, and the second derivative

For $f(x) = -x^2 + 6x$, determine the interval where f is decreasing.

$f'(x) = -2x + 6$. Decreasing when $f'(x) < 0$, i.e. $-2x + 6 < 0 \Rightarrow x > 3$.

10 [Intermediate] • Target: Increasing, decreasing, and the second derivative

Find the second derivative of $f(x) = x^4 - 2x^2$.

$f'(x) = 4x^3 - 4x$, so $f''(x) = 12x^2 - 4$.

11 [Intermediate] • Target: Word problems: rates of change

This question has two parts. The height of a ball thrown upward is given by $h(t) = -5t^2 + 20t + 2$ (in meters, t in seconds).

a Find the velocity function $h'(t)$.

$h'(t) = -10t + 20$.

b Find the velocity at $t = 1$ second, and state whether the ball is rising or falling.

$h'(1) = -10(1) + 20 = 10$ m/s, which is positive, so the ball is rising.

(a) $h'(t) = -10t + 20$.; (b) $h'(1) = -10(1) + 20 = 10$ m/s, which is positive, so the ball is rising.

12 [Intermediate] • Target: Word problems: rates of change

This question has two parts. A company's cost to produce x items is $C(x) = 0.01x^2 + 5x + 200$ (in dollars).

a Find the marginal cost function $C'(x)$.

$C'(x) = 0.02x + 5$.

b Find the marginal cost at $x = 100$ items, and interpret its meaning.

$C'(100) = 0.02(100) + 5 = 2 + 5 = 7$ dollars — approximately the cost of producing one more item beyond the 100th.

(a) $C'(x) = 0.02x + 5$.; (b) $C'(100) = 0.02(100) + 5 = 2 + 5 = 7$ dollars — approximately the cost of producing one more item beyond the 100th.

13 [Intermediate] • Target: MCQ — Evaluating limits by factoring (the 0/0 form)

Differentiate the function $f(x) = (2x + 1)^4$ using the Chain Rule.

Step 1: Apply Chain Rule: $\frac{d}{dx}[u(x)^n] = nu(x)^{n-1} \cdot u'(x)$. Step 2: Differentiate: Here $u(x) = 2x + 1$ and $u'(x) = 2$: $f'(x) = 4(2x + 1)^3 \cdot (2) = \mathbf{8(2x + 1)^3}$.

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14 [Intermediate] • Target: MCQ — Limits at infinity

Evaluate the limit at infinity: $\lim_{x\to\infty} \frac{6x^2-5}{2x^2+7x}$

Step 1: Compare degrees of polynomials: Both numerator and denominator have degree 2. Step 2: Apply the leading coefficient rule: $\lim_{x\to\infty} \frac{6x^2-5}{2x^2+7x} = \frac{6}{2} = \mathbf{3}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: MCQ — One-sided limits • Cross-Lesson: 2.01 Functions & Transformations

Find the slope of the tangent line to the curve $y = 2x^3 - 5x + 1$ at $x = 2$.

Step 1: Differentiate using the power rule: $y' = \frac{d}{dx}(2x^3 - 5x + 1) = 6x^2 - 5$. Step 2: Evaluate at $x = 2$: $m = 6(2)^2 - 5 = 6(4) - 5 = 24 - 5 = \mathbf{19}$.

16 [Challenging] • Target: MCQ — The power rule • Cross-Lesson: 3.01 Sequences & Series

Level 1 · FoundationsFind the derivative of $y = \sqrt{x^4 + 3x^2 + 7}$ using the Chain Rule, and compute $y'(1)$.

Express $y = (x^4 + 3x^2 + 7)^{1/2}$. By the Chain Rule: $y' = \frac{1}{2}(x^4 + 3x^2 + 7)^{-1/2} \cdot \frac{d}{dx}(x^4 + 3x^2 + 7) = \frac{4x^3 + 6x}{2\sqrt{x^4 + 3x^2 + 7}} = \frac{2x^3 + 3x}{\sqrt{x^4 + 3x^2 + 7}}$. Evaluate at $x = 1$: $y'(1) = \frac{2(1)^3 + 3(1)}{\sqrt{1^4 + 3(1)^2 + 7}} = \frac{5}{\sqrt{11}} = \mathbf{\frac{5\sqrt{11}}{11}}$.

17 [Challenging] • Target: MCQ — The product rule • Cross-Lesson: 4.01 Trigonometry

Given the functions $f(x) = x^2 - 5x + 6$ and $g(x) = |x - 2|$, find the value of $f(g(4))$.

To find $f(g(4))$, we first calculate the inner value $g(4)$. The function $g(x) = |x - 2|$, so $g(4) = |4 - 2| = |2| = 2$. Now, we substitute the value of $g(4)$ into the function $f(x)$. We need to calculate $f(2)$. The function $f(x) = x^2 - 5x + 6$, so $f(2) = (2)^2 - 5(2) + 6 = 4 - 10 + 6 = 0$. Therefore, $f(g(4)) = 0$.

18 [Challenging] • Target: MCQ — The chain rule • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

What is the measure of each interior angle in a regular octagon (8-sided polygon)?

Step 1: Compute exterior angle: $\theta_{\text{ext}} = \frac{360^\circ}{n} = \frac{360^\circ}{8} = 45^\circ$. Step 2: Subtract from 180°: $\theta_{\text{int}} = 180^\circ - 45^\circ = \mathbf{135}^\circ$.

19 [Challenging] • Target: MCQ — Derivatives of trigonometric and exponential functions • Cross-Lesson: 7.01 Integration & Applications

Find the slope of the tangent to $y = x^2$ at $x = 2$.

Step 1: Compute first derivative: The derivative represents the slope function $m(x)$: $y' = \frac{d}{dx}[x^2] = 2x$ Step 2: Evaluate at $x = 2$: $m = y'(2) = 2(2) = \mathbf{4}$

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20 [Challenging] • Target: MCQ — Tangent lines • Cross-Lesson: 8.01 Exponentials & Logarithms

Find the area bounded by the curve $y = 3x^2$, the x-axis, and the vertical lines $x = 1$ and $x = 2$.

Step 1: Set up the area integral: $A = \int_1^2 3x^2\,dx$. Step 2: Integrate and substitute: $A = [x^3]_1^2 = 2^3 - 1^3 = 8 - 1 = \mathbf{7}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — One-sided limits

Using the same piecewise function, find $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.

a. 4**; does not exist** ✓

b. 4; exists

c. 2; exists

d. 2; does not exist

Right limit $= 1 + 3 = 4$. Since the left limit (2) differs from the right limit (4), the two-sided limit does not exist.

22 [Foundational] • Target: MCQ — Tangent lines

The graph of $f(x) = x^2 - 2x$ is shown, with the point at $x = 3$ marked. Find the equation of the tangent line there.

a. $y = 3x - 9$

b. $y = 4x - 3$

c. $y = 4x + 9$

d. $y = 4x - 9$ ✓

$f(3) = 3$, $f'(3) = 4$: $y - 3 = 4(x - 3) \Rightarrow y = 4x - 9$.

23 [Foundational] • Target: MCQ — Tangent lines

The graph shows $f(x) = x^2$ and its tangent line at $x = 2$. Find the slope of the tangent line by reading the graph.

a. 4 ✓

b. 2

c. 1

d. 8

The tangent line rises 4 for every 1 across.

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24 [Foundational] • Target: MCQ — Word problems: rates of change

The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the ball's velocity at $t = 2$ seconds.

a. 10 ✓

b. -10

c. 30

d. 20

$h'(t) = -10t + 30$; $h'(2) = 10$ m/s.

25 [Foundational] • Target: MCQ — Word problems: rates of change

A company's cost to produce x items is $C(x) = 0.02x^2 + 8x + 150$ (in dollars). Find the marginal cost at $x = 100$.

a. 8 dollars

b. 4 dollars

c. 20 dollars

d. 12 **dollars** ✓

$C'(x) = 0.04x + 8$; $C'(100) = 12$.

26 [Foundational] • Target: MCQ — Word problems: rates of change

The height of a ball thrown upward is given by $h(t) = -5t^2 + 30t + 3$ (in meters). Find the time at which the ball reaches its maximum height.

a. $t = 3$ ✓

b. $t = 5$

c. $t = 6$

d. $t = 2$

$h'(t) = -10t + 30 = 0 \Rightarrow t = 3$ seconds.

27 [Foundational] • Target: The chain rule

For $f(x) = -x^2 - 3$, determine the interval where f is increasing.

a. $x < 0$ ✓

b. $x > 2$

c. $x < 2$

d. $x > 0$

$f'(x) = -2x$. Increasing when $f'(x) > 0$, i.e. $x < 0$.

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28 [Foundational] • Target: Derivatives of trigonometric and exponential functions

■■ ■■■■■■ ■■■■■ ■■■■■■ ■■■■■■■ $f(x) = x^2 \sin x$■ Product rule derivative of $x^2 \sin x$?

a. $2x\sin x + x^2\cos x$ ✓

b. $2x\cos x$

c. $x^2\cos x - 2x\sin x$

d. $2x\sin x$

■■■■■ ■■■■■: $f'(x) = (x^2)'\sin x + x^2(\sin x)' = 2x\sin x + x^2\cos x$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Tangent lines

■■ ■■■■■■ ■■■■■■■ ■■■■■■■■ $f(x) = \frac{x}{x+1}$■ Quotient rule derivative of $\frac{x}{x+1}$?

a. $\frac{1}{(x+1)^2}$ ✓

b. $\frac{2x+1}{(x+1)^2}$

c. $\frac{-1}{(x+1)^2}$

d. 1

■■■■■ ■■■■■■■: $\frac{(1)(x+1)-(x)(1)}{(x+1)^2} = \frac{x+1-x}{(x+1)^2} = \frac{1}{(x+1)^2}$.

30 [Intermediate] • Target: Increasing, decreasing, and the second derivative

■■ ■■■■■ ■■■■■■■■ $\lim_{x\to 2}(3x^2 - 4x + 1)$■ Evaluate limit by direct substitution.

a. **5** ✓

b. 4

c. 7

d. 1

■■■■■■■■ ■■■■■■■■: $3(2^2) - 4(2) + 1 = 3(4) - 8 + 1 = 12 - 8 + 1 = 5$.

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31 **[Intermediate]** • **Target: Word problems: rates of change**

■■■ ■■■■ $f(x) = x^3 - 6x^2 + 5$ ■■■■■ ■■■■ $g(x)$ ■■■ ■■■ ■■■■■■■■ ■■■■■■■ ■■■ $x = 4$ ■■■ $f(x)$■ ■■■■■■■ $g(4) = f(4)$■ ■■■ ■■ ■■■■ $g'(4)$■ If $f(x) = x^3 - 6x^2 + 5$ and another function $g(x)$ has the same first derivative at $x = 4$ as $f(x)$, and also $g(4) = f(4)$, what is the value of $g'(4)$?

a. 11

b. -19

c. 5

d. **-7** ✓

■■■■■■■ ■■■■ $g'(4)$■ ■■■■■■ ■■■■■■ ■■■ ■■■■ ■■■■■■■■ ■■■■■■■ ■■■■■■■ $f(x)$ ■■ ■■■■■■■■ ■■■ $x = 4$. ■■■■■■ $f(x) = x^3 - 6x^2 + 5$. ■■■■■■ $f(x)$ ■■ $f'(x) = \frac{d}{dx}(x^3 - 6x^2 + 5) = 3x^2 - 12x$. ■■■■■■ ■■■■■ ■■■■■■■ $f'(x)$ ■■■ $x = 4$: $f'(4) = 3(4)^2 - 12(4) = 3(16) - 48 = 48 - 48 = 0$. ■■■■■■ ■■■■■■■■■ $g(x)$ ■■■ ■■■ ■■■■■■■■ ■■■■■■■ ■■■ $x = 4$ ■■■ $f(x)$. ■■■ ■■■■■ ■■ $g'(4) = f'(4)$. ■■■■■■ $g'(4) = 0$. ■■■■■ ■■ ■■■■■■ $g(4) = f(4)$ ■■■ ■■■■■■ ■■■■■■ $g'(4)$ ■■■■■ ■■ ■■■■■ ■■■■■■■ ■■ ■■■■■■ ■■■■■ ■■■■■■ ■■ $g(x)$. To find the value o.

32 **[Intermediate]** • **Target: MCQ — Evaluating limits by direct substitution**

■■■ ■■■ ■■■■■ ■■■■■■ ■■■■■■■ $f(x) = x^3 - 3x$ ■■■■■■■■ ■■■ ■■ ■■■ x ■■■■■■■■■ Find x where tangent line is horizontal.

a. ± 1 ✓

b. 0 ■■■

c. $\pm\sqrt{3}$

d. 3 ■■■

■■■■■■■ ■■■■■■■ ■■■■■ ■■ ■■■■■■ ■■■: $f'(x) = 3x^2 - 3 = 0 \Rightarrow 3(x^2 - 1) = 0 \Rightarrow x = \pm 1$.

33 **[Intermediate]** • **Target: MCQ — Evaluating limits by factoring (the 0/0 form)**

■■ ■■■■■■■■ ■■■■■ ■■■■■ ■■■■■ ■■■■■■■ $f(x) = x^3 - 3x$ ■■■■■■■■■ Intervals where $x^3 - 3x$ is increasing?

a. $(-\infty, -1) \cup (1, \infty)$ ✓

b. $(-1, 1)$

c. $(0, \infty)$

d. $(-\infty, 0)$

$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1) > 0$. ■■■■■■■■ ■■■■■■ ■■■■■ ■■■■■■■■■: $(-\infty, -1) \cup (1, \infty)$.

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34 [Intermediate] • Target: MCQ — Limits at infinity

■■ ■■■■■ ■■■■■■ $G(x) = \int_0^{x^2} \sin(t)\,dt$ ■■■■■■■ ■■■ x■ Derivative using Leibniz rule / chain rule?

a. $2x\sin(x^2)$ ✓

b. $\sin(x^2)$

c. $\cos(x^2)$

d. $2x\cos(x^2)$

■■■■■■ ■■■■■■■ ■■■■■■■ ■■■■■■■■■: $G'(x) = \sin(x^2) \times \frac{d}{dx}(x^2) = 2x\sin(x^2)$.

35 [Intermediate] • Target: MCQ — One-sided limits

■■ ■■■■■ ■■■■■■ ■■■■■■■ $f(x) = 15$■ Derivative of constant function $f(x) = 15$?

a. **0** ✓

b. 1

c. 15

d. x

■■■■■ ■■ ■■■■ ■■■■■ ■■■■■ ■■■■■ ■■■■■■: $f'(x) = 0$.

36 [Intermediate] • Target: MCQ — The power rule

■■ ■■■■■ ■■■■■■ $f(x) = 7x^4$■ Derivative of $f(x) = 7x^4$?

a. $28x^3$ ✓

b. $21x^3$

c. $28x^4$

d. $7x^3$

■■■■■ ■■■■■: $f'(x) = 7(4)x^3 = 28x^3$.

37 [Intermediate] • Target: MCQ — The product rule

■■■■ ■■■■■ ■■■■■■ $f(x) = \frac{x^3-8}{x-2}$. ■■■ ■■■■ $f(2)$ ■■■■ ■■■■ ■■■■■■ $f(x)$ ■■■■■ ■■■ $x = 2$■ ■■■ ■■ ■■■■ $f(2)$■ Consider the function $f(x) = \frac{x^3-8}{x-2}$. If $f(2)$ is defined such that the function $f(x)$ is continuous at $x = 2$, what is the value of $f(2)$?

a. 12 ✓

b. 10

c. 8

d. 6

■■■ ■■■■ ■■■■■■ $f(x)$ ■■■■■ ■■■ $x = 2$■ ■■■ ■■ ■■■■ $\lim_{x\to 2} f(x) = f(2)$. ■■■■■ ■■■■■ ■■■ ■■■■ ■■■■■■■■: $\lim_{x\to 2} \frac{x^3-8}{x-2}$. ■■■ ■■■■■■■■ ■■■■■■■■ ■■ $x = 2$■ ■■■■ ■■■ ■■■■■■■ ■■■ ■■■■■■■■ $\frac{0}{0}$■ ■■■■ ■■■■■ ■■■ ■■■■■■■ ■■■■■■ ■■■■■■. ■■■■■ ■■■■■■ ■■■■■■ $x^3 - 8$ ■■■■■ ■■■ ■■■■■■■: $x^3 - 2^3 = (x-2)(x^2+2x+4)$. ■■■■■ $f(x) = \frac{(x-2)(x^2+2x+4)}{x-2}$. ■■■■■■■■ ■■ $xeq2$■ ■■■■■■■ ■■■■■■■ $(x-2)$ ■■ ■■■■■■ ■■■■■■■■..

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: MCQ — The chain rule** • **Cross-Lesson: 5.01 Geometry & Coordinate Geometry**

[redacted] $f(x) = x^2 - 4x + 3$. [redacted] x [redacted] 3. Let the function be $f(x) = x^2 - 4x + 3$. Find the limit of the function as x approaches 3.

a. 0 ✓

b. 3

c. 6

d. 9

[redacted] $f(x) = x^2 - 4x + 3$ [redacted] x [redacted] 3[redacted] $x = 3$ [redacted] [redacted] $x = 3$. \lim_{x \to 3} (x^2 - 4x + 3) \) $= (3)^2 - 4(3) + 3 = 9 - 12 + 3 = 0$ To find the limit of the function $f(x) = x^2 - 4x + 3$ as x approaches 3, we directly substitute $x = 3$ into the function since it is a polynomial function and continuous at $x = 3$. \lim_{x \to 3} (x^2 - 4x + 3) \) $= (3)^2 - 4(3) + 3 = 9 - 12 + 3 = 0$

39 **[Challenging]** • **Target: MCQ — Derivatives of trigonometric and exponential functions** • **Cross-Lesson: 7.01 Integration & Applications**

[redacted] $f(x) = x^2 - 3$[redacted] [redacted] x [redacted] If $f(x) = x^2 - 3$, find the critical point and classify it.

a. x=0 (local maximum)

b. x=1 (local minimum)

c. x=-1 (local minimum)

d. **x=0 (local minimum)** ✓

$f'(x) = 2(1)x = 0 \Rightarrow x = 0; a > 0 \Rightarrow$ local minimum

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40 [Challenging] • Target: MCQ — Tangent lines • Cross-Lesson: 8.01 Exponentials & Logarithms

[illegible] $f(x)=\frac{x^2+ax+4}{x-2}$[illegible] [illegible] a [illegible] [illegible]. [illegible] [illegible] $\lim_{x\to 2} f(x)$ [illegible] [illegible] [illegible] [illegible] a[illegible] Given the function $f(x)=\frac{x^2+ax+4}{x-2}$, where a is a constant. If $\lim_{x\to 2} f(x)$ exists, what is the value of a?

a. **-4** ✓

b. 2

c. 4

d. -2

[illegible] [illegible] [illegible] $\lim_{x\to 2} f(x)$ [illegible] [illegible] [illegible] $f(x)=\frac{x^2+ax+4}{x-2}$[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $x=2$. [illegible] [illegible] [illegible] $(x-2)$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] $x=2$ [illegible] [illegible] [illegible] [illegible]: $2^2+a(2)+4=0$ $4+2a+4=0$ $2a+8=0$ $2a=-8$ $a=-4$ [illegible] $a=-4$[illegible] [illegible] [illegible] $f(x)=\frac{x^2-4x+4}{x-2}=\frac{(x-2)^2}{x-2}=x-2$ ([illegible] $x\neq 2$).

[illegible] $\lim_{x\to 2}(x-2)=2-2=0$. For the l.

41 [Challenging] • Target: MCQ — Increasing, decreasing, and the second derivative • Cross-Lesson: 9.01 Statistics & Probability

[illegible] [illegible] [illegible] $f(x)=3x^2+5x-2$ [illegible] [illegible] x [illegible] 1[illegible] [illegible] [illegible] [illegible] To find the limit of the function $f(x)=3x^2+5x-2$ as x approaches 1, what is the value?

a. **10** ✓

b. 6

c. 7

d. 9

[illegible] [illegible] [illegible] $f(x)=3x^2+5x-2$ [illegible] [illegible] x [illegible] 1[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $x=1$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. $\lim_{x\to 1}(3x^2+5x-2)=3(1)^2+5(1)-2$ $=3(1)+5-2$ $=3+5-2$ $=8-2$ $=6$ To find the limit of the function $f(x)=3x^2+5x-2$ as x approaches 1, we substitute $x=1$ directly into the function because the function is a polynomial and continuous at all points.

$\lim_{x\to 1}(3x^2+5x-2)=3(1)^2+5(1)-2$.

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42 [Challenging] • Target: MCQ — Word problems: rates of change • Cross-Lesson: 1.01 Algebra & Equations

[redacted] $f(x) = \frac{ax^2 - 5x + 3}{x - 1}$ [redacted] $x = 1$[redacted] [redacted] a. If the function $f(x) = \frac{ax^2 - 5x + 3}{x - 1}$ has a limit as $x \to 1$, find the value of a.

a. 1

b. **2** ✓

c. 3

d. 5

[redacted] $f(x) = \frac{ax^2 - 5x + 3}{x - 1}$ [redacted] $x = 1$[redacted] [redacted] [redacted] [redacted] [redacted] [redacted] $x = 1$ [redacted]. [redacted] $x = 1$ [redacted]: $a(1)^2 - 5(1) + 3 = 0$ $a - 5 + 3 = 0$ $a - 2 = 0$ $a = 2$. For the function $f(x) = \frac{ax^2 - 5x + 3}{x - 1}$ to have a limit as $x \to 1$, the numerator must be zero at $x = 1$ to avoid an indeterminate form of division by zero. Substituting $x = 1$ into the numerator: $a(1)^2 - 5(1) + 3 = 0$ $a - 5 + 3 = 0$ $a - 2 = 0$ $a = 2$.

43 [Challenging] • Target: Evaluating limits by direct substitution • Cross-Lesson: 2.01 Functions & Transformations

[redacted] $f(x) = 3x + 5$ [redacted] x [redacted] 2 [redacted]:\ The limit of the function $f(x) = 3x + 5$ as x approaches 2 is:

a. 8

b. 9

c. **11** ✓

d. 13

[redacted] x [redacted]:

$\lim_{x \to 2} (3x + 5) = 3(2) + 5 = 6 + 5 = 11$ Therefore, the limit is 11. To find the limit directly, substitute the value of x into the function: $\lim_{x \to 2} (3x + 5) = 3(2) + 5 = 6 + 5 = 11$ Therefore, the limit is 11.

44 [Challenging] • Target: Evaluating limits by factoring (the 0/0 form) • Cross-Lesson: 3.01 Sequences & Series

[redacted] $f(x) = -x^2 + 4x$: Local max of $f(x) = -x^2 + 4x$.

a. $x = 1$

b. $x = 2$ ✓

c. $x = 4$

d. $x = 0$

[redacted] Limits & Derivatives: $f'(x) = -2x + 4 = 0 \Rightarrow x = 2$ [redacted] [redacted] [redacted] $x = 2$. Applying Limits & Derivatives principles: $f'(x) = -2x + 4 = 0 \Rightarrow x = 2$ Therefore, the correct choice is $x = 2$.

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45 [Challenging] • Target: Limits at infinity • Cross-Lesson: 4.01 Trigonometry

■■ ■■■■ ■■■■■■■ $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$■ Evaluate limit involving radical.

a. $\frac{1}{4}$ ✓

b. $\frac{1}{2}$

c. 0

d. 4

■■■■■■ ■■ ■■■■■■ ■■■■■■ $\sqrt{x}+2$: $\frac{x-4}{(x-4)(\sqrt{x}+2)} = \frac{1}{\sqrt{x}+2}$. ■■■■■■■■■: $\frac{1}{2+2} = \frac{1}{4}$.

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